

BANARAS HINDU UNIVERSITY

Sessional Test, Sem-III, 2025-26

Subject: Statistics

Paper: STAMV-32

Time: 2 hours

Total Marks: 20

*Note: Attempt any **Two** questions. Question 1 is compulsory.*

1. Draw 1,000 fill () sequence of numbers from -3 to $+3$ in MS Excel. Now, Calculate the value of function $f(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ by treating each previous value as x . Now, plot $x/f(x)$ on x-y graph of excel. Now, calculate another function $f(x) = \frac{1}{\sqrt{2\pi}}e^{-(x-2)^2/2}$. Insert the new x & $f(x)$ in same graph. Explain your findings.

2. A website logs number of server errors per hour for 40 hours:

Errors/hour (x)	0	1	2	3	4
Observed frequency	10	13	9	5	3

- (a) Estimate λ .
 - (b) Fit Poisson model.
 - (c) Compute expected frequencies.
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3. The marks of 120 students in a statistics test are:

Marks	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Observed frequency	8	15	22	30	24	14	7

- (a) Estimate μ and σ .
 - (b) Fit Normal distribution.
 - (c) Compute expected frequencies.
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